

DTU



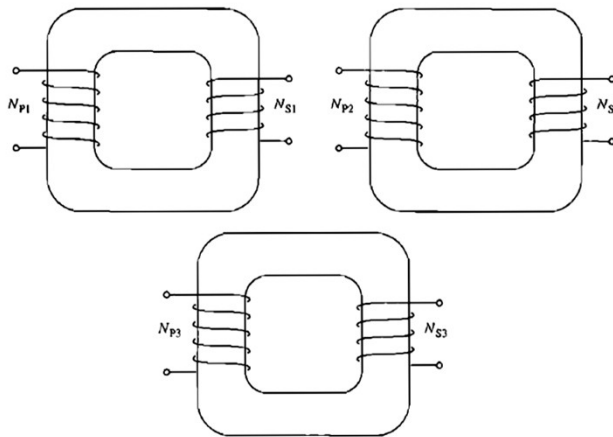
# Transformers

## **62768 Electrical energy systems, project**

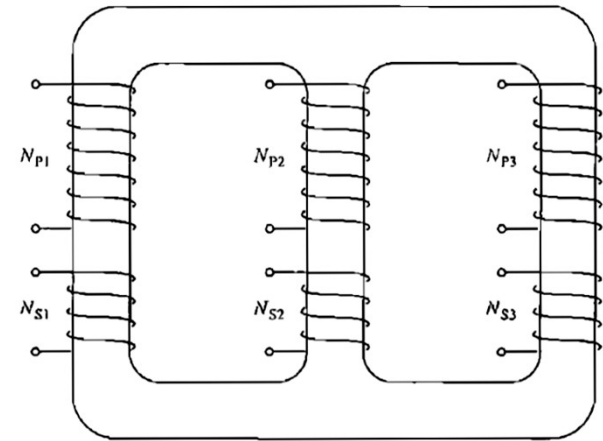
Ashraf Khalil  
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# THREE-PHASE TRANSFORMERS

- A single three-phase transformer is lighter, smaller, cheaper, and slightly more efficient, but using three separate single-phase transformers has the advantage that each unit in the bank could be replaced individually in the event of trouble.
- A utility would only need to stock a single spare single-phase transformer to back up all three phases, potentially saving money.



Three single-phase transformers



Three sets of windings wrapped on a common core.

## Three-Phase Transformer Connections

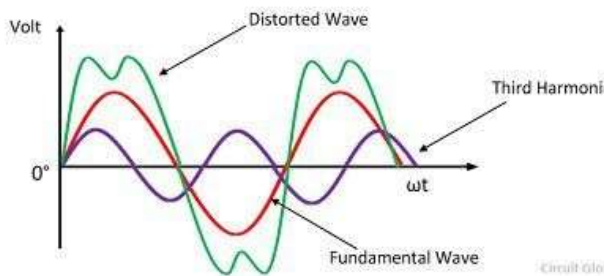
- 1. Wye- wye ( $Y - Y$ )
  - 2. Wye-delta ( $Y - \Delta$ )
  - 3. Delta-wye ( $\Delta - Y$ )
  - 4. Delta-delta ( $\Delta - \Delta$ )
- 
- Any single transformer in the bank behaves exactly like the single-phase transformers already studied.
  - The impedance, voltage regulation, efficiency, and similar calculations for three-phase transformers are done on a per-phase basis, using exactly the same techniques already developed for single-phase transformers.

# WYE-WYE CONNECTION

The Y - Y connection has two very serious problems;

1. If loads on the transformer circuit are unbalanced, then the voltages on the phases of the transformer can become severely unbalanced.
2. Third-harmonic voltages can be large.

$$\frac{V_{LP}}{V_{LS}} = \frac{\sqrt{3}V_{\phi P}}{\sqrt{3}V_{\phi S}} = a$$



The third-harmonic components of each of the three phases will be in phase with each other.

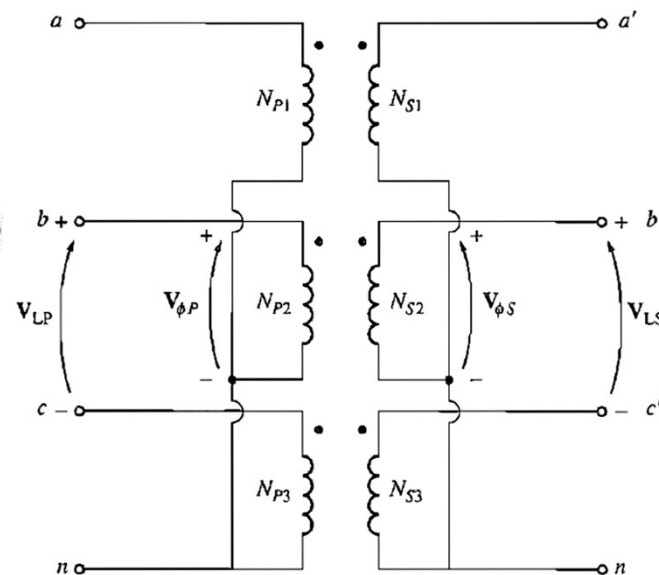
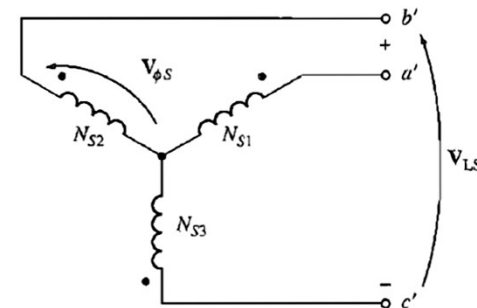
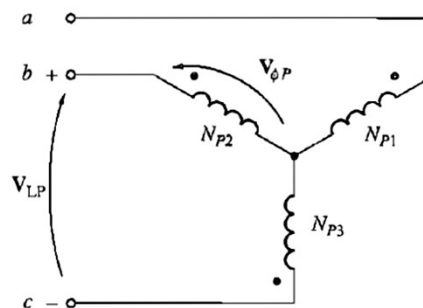
$$I_{AO} = I_{1m} \sin \omega t + I_{3m} \sin(3\omega t + \phi_3) + I_{5m} \sin(5\omega t + \phi_5) \dots \dots \text{equ}(1)$$

$$I_{BO} = I_{1m} \sin(\omega t - 120^\circ) + I_{3m} \sin[3(\omega t - 120^\circ) + \phi_3] + I_{5m} \sin[3(\omega t - 120^\circ) + \phi_3]$$

$$I_{BO} = I_{1m} \sin(\omega t - 120^\circ) + I_{3m} \sin(3\omega t + \phi_3) + I_{5m} \sin(5(\omega t + 120^\circ) + \phi_5) \dots \dots \text{equ}(2)$$

$$I_{CO} = I_{1m} \sin(\omega t - 240^\circ) + I_{3m} \sin[3(\omega t - 240^\circ) + \phi_3] + I_{5m} \sin[3(\omega t - 240^\circ) + \phi_3]$$

$$I_{CO} = I_{1m} \sin(\omega t - 240^\circ) + I_{3m} \sin(3\omega t + \phi_3) + I_{5m} \sin(5(\omega t + 240^\circ) + \phi_5) \dots \dots \text{equ}(3)$$



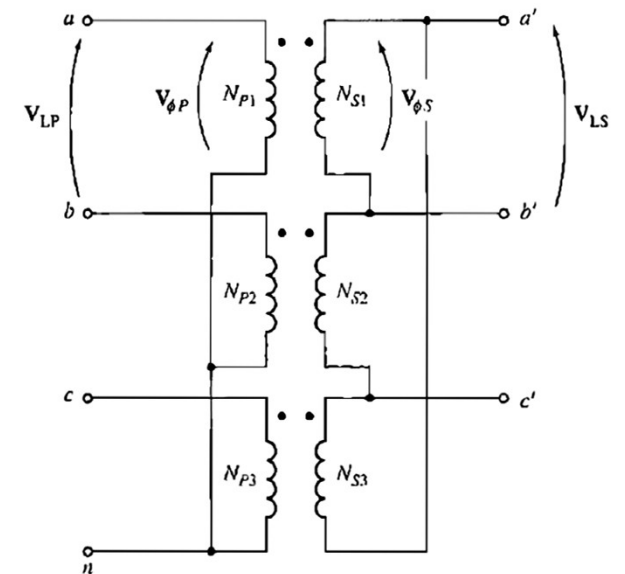
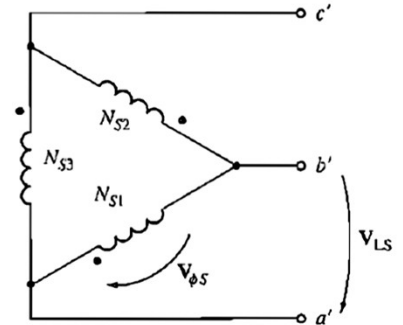
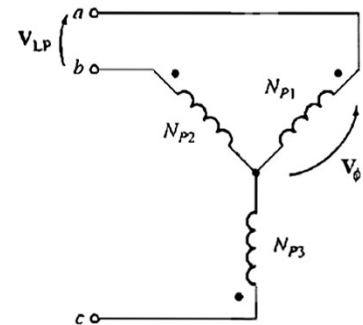
# WYE-DELTA CONNECTION

The Y -  $\Delta$  connection has no problem with third-harmonic components in its voltages, since they are consumed in a circulating current on the a side. This connection is also more stable with respect to unbalanced loads, since the a partially redistributes any imbalance that occurs.

$$\frac{V_{\phi P}}{V_{\phi S}} = a$$

$$\frac{V_{LP}}{V_{LS}} = \frac{\sqrt{3}V_{\phi P}}{V_{\phi S}}$$

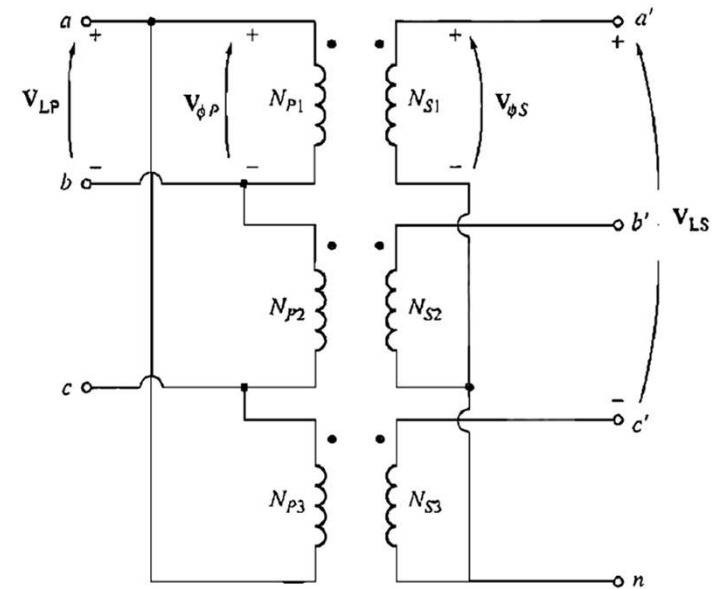
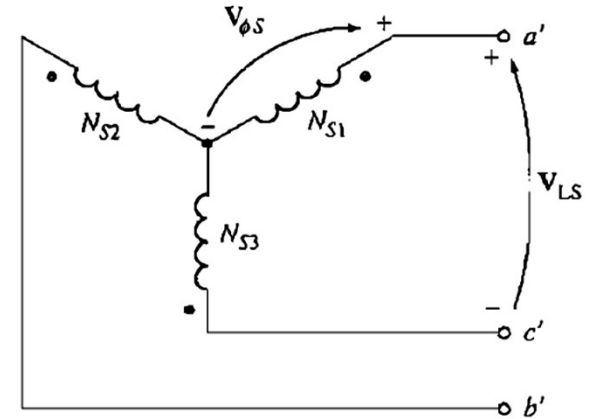
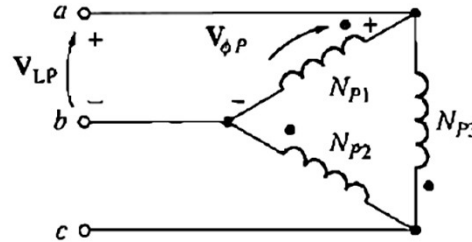
$$\frac{V_{LP}}{V_{LS}} = \sqrt{3} a$$



# DELTA-WYE CONNECTION

$$\frac{V_{LP}}{V_{LS}} = \frac{V_{\phi P}}{\sqrt{3}V_{\phi S}}$$

$$\frac{V_{LP}}{V_{LS}} = \frac{a}{\sqrt{3}}$$



# DELTA-DELTA CONNECTION

This transformer has no phase shift associated with it and no problems with unbalanced loads or harmonics

$$\frac{V_{LP}}{V_{LS}} = \frac{V_{\phi P}}{V_{\phi S}} = a$$

